Ethan Soh

Applied AI · Agent Engineering · AI Security

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Looking for a 44-week internship from 12 April 2027 to 11 February 2028.

ABOUT ME

I'm a second-year Cybersecurity and Digital Forensics student at Singapore Polytechnic. Most of my time outside class goes into building things with AI agents and then trying to make them behave: I write the rules they follow, the checks that stop them, and the tests that prove the checks work. I hold Anthropic's Claude Certified Architect – Foundations (CCA-F) and ISC2 Certified in Cybersecurity. I'm looking for an internship in applied AI, agent engineering or AI security.

WHAT I'VE BUILT

Unburden Root · Personal project — Jul 2026 – present

- The setup I run all my other work on. Claude Code sessions act as named roles (an orchestrator, reviewers, one session per project) and share memory and a set of written rules. 43 hook scripts enforce those rules, a router sends each job to the cheapest model that can do it, and a guard that handles personal data refuses when it can't tell whether something is private. I direct the agents and review what they write; I don't pretend I typed every line.

Agentroid · Own repository, public release planned — Oct 2026 – present

- An MCP server that lets an AI agent operate Android emulators. The agent doing the calling does the thinking, so Agentroid needs no model API key of its own. What I care about most is the safety model: every adb command is blocked unless allowed, each device has to be on an allowlist, every action carries a risk label from R0 to R5, and sessions can be recorded and replayed. The core is Rust, with a TypeScript MCP server and a Python SDK, written with AI coding agents under my design and review. It drives emulators only, and the README lists what it cannot do.

Awards and attendance system, The Boys' Brigade 5th Singapore Company — Apr 2026 – present

- Started as a Google Sheets tracker for the awards department (live 27 Apr 2026); from July, a Next.js, React and Supabase portal. It holds awards and attendance records and imports the training programme. I also built an AI assistant that only ever saw pseudonymised names and needed a person to confirm every write; I took it out in October after reviewing it, because the portal did not need that risk. We applied to the National Youth Council's Young ChangeMakers grant in September.

NUS-ISS Show Me Your Agents Hackathon 2026 · Team of four, with AWS and the Singapore Business Federation — Sep – Oct 2026

- We built an agentic AI solution to a real SME problem. We were not among the teams selected; I have the Letter of Participation.

SMALL PUBLIC REPOS

- **detached-run** zero-dependency Node tool that starts Windows background processes through WMI so they outlive their parent app and never open a window; its test suite checks that no window appears.
- **reddit-pull** zero-dependency Node tool that pulls public Reddit data through the Arctic Shift archive and prints one JSON line per item, so an AI agent can read it; tested on Linux, Windows and macOS.
- **Sidemic** turns an Android phone into a Windows PC's microphone, with noise removal on the phone and a Rust and Tauri desktop app.
- **designyourtv** a painted depth-clock lock screen, glass widgets and a remote-driven dock for Google TV.
- **android-film-simulations** a phone-and-monitor colour rig (adb, ArUco, CIEDE2000) that measured 11 Fujifilm looks from about 490 shots; best in-camera settings cut Acros error from 25.6 to 2.1 ΔE, an exact .cube tier reaches 0.2 to 0.4.

Also in progress: Claude for HMOS Lite, so I can answer my coding agent from a Huawei watch (unofficial, not affiliated with Anthropic or Huawei).

WORK

Academic Content Assistant · Alchemy Education Studio · contract, remote — Feb 2025 – Oct 2026

- I write O-Level Biology material for a tuition centre: worksheets, revision papers, mock exams, answer keys and marking schemes, each checked for accuracy, syllabus fit and marking-point detail. I finished every paper for the 6093 Biology syllabus (Sec 3 and 4), and the centre uses them in lessons. I also built a pipeline that drafts each paper from a reference library and checks it.

Retail Assistant · Bee Cheng Hiang · part-time — Nov 2024 – Feb 2025

- Chinese New Year season: till, packing, restocking and customer questions, including Mandarin.

CERTIFICATIONS

Claude Certified Architect – Foundations (CCA-F)
Anthropic · issued Sep 2026, valid to Sep 2027
View on Credly

Certified in Cybersecurity (CC)
ISC2 · issued Jan 2026, valid to Jan 2029
View on Credly

Anthropic Academy, five courses (Claude 101, Claude Code 101, Claude Platform 101, Cowork, AI Fluency), Jul 2026. CPR + AED Provider, Singapore Red Cross, Oct 2026. Anthropic Cyber Verification Program, approved Jul 2026.

SCHOOLWORK I'M PROUD OF

- **Fortune's Descent** My Back-End Web Development CA2: a browser game on an Express API with JWT sign-in, validation and tests. I went through the code myself and can explain every line.
- **Offensive Security CA2** Group project: we assessed CVE-2025-47812 (Wing FTP Server) and scored it with CVSS, using Claude Pro.
- **Securing Microsoft Windows CA1** I did almost all of the hands-on Windows Server 2022 hardening, with Claude guiding: group policy, Active Directory, DNS, CIS audits.

COMPETITIONS AND EVENTS

- **Hack The Box Cyber Apocalypse CTF 2026** 30th of 6,744 teams, with Team Mythos Larpers. We solved all 136 challenges for 69,425 points.
- **Singapore Cyber Conquest 2026, Student Category** one of the top 20 teams from the online qualifiers; I went to the finals at Singapore International Cyber Week on 14 Oct.
- **DSTA BrainHack 2026, Cyber Defenders Discovery Camp** finalist in the Explorer category with team TUFF.
- **CyberBlitz 2026** I wrote binary-exploitation challenges for SP Gryphons, each with a Docker setup, a solver and a writeup.
- **National AI Student Challenge 2026, Workato track** participant, in a team of four. The brief was an MCP prototype.
- **TISC 2026 (CSIT)** Participant.

Also went to: Black Hat Asia 2026 (student pass), OB1 Conference 2026 (selected for SP's student sponsorship with STAR Labs SG), DEF CON Singapore 2026 (represented SP Gryphons), the OffSec Singapore Chapter meetup (May 2026).

LEADERSHIP AND VOLUNTEERING

Senior Cadet Lieutenant (Primer) · The Boys' Brigade 5th Singapore Company — Dec 2024 – present

- I'm the link between the Boys and the Officers. I'm the Company's only Drill Instructor in 2026 (I passed both the practical and theory assessments). I was second-in-command of Blaze 2026, where I trained three teams for an interschool 21 km adventure race: first aid, knots, orienteering and fitness. I was Assistant Camp Commandant for Adventure Camp 2026 and ran the day-to-day operations, and I'm third-in-command of the Awards Department.

Senior Exco member, SEED (Computer Science Club) · Singapore Polytechnic — 2026

- I organised and taught the "How to COMe Home" Computer Organisation crash course for juniors on 20 May 2026.

Publicity Advisor, Gryphons (Cybersecurity Club) · Singapore Polytechnic — 2026

- I designed the poster for the joint SEED and Gryphons AGM and helped at the Youth Cyber Exploration Camp.

Class Management Committee · Singapore Polytechnic — 2026

- Planned the class outing and passed information between teachers and the class.

AWARDS

- Founder's Award, The Boys' Brigade in Singapore, Jun 2024. It is the highest award in the Senior Programme.
- Best Primer Merit Award, 5th Company, 4 Apr 2026.
- Edusave Certificate of Academic Achievement 2026 (top 25% of my level and course); MOE EAGLES Award and Edusave Good Progress Award, Nov 2024.
- Black Belt, Dan 1, Singapore Taekwondo Federation.

EDUCATION

Singapore Polytechnic · Diploma in Cybersecurity and Digital Forensics — Apr 2025 – Mar 2028

Distinctions in Digital Forensics and Investigation and Personal Branding and Career Agility. Other modules: Ethical Hacking Essentials, Offensive Security, Linux Administration and Security, Securing Microsoft Windows, Back-End and Front-End Web Development, Programming Methodology, Computer Organisation, AI for Impact. This semester: Malware Reverse Engineering, Applied Cryptography, Data Protection for Cyber Security.

Kuo Chuan Presbyterian Secondary School · GCE O-Level, L1R4: 9 — Jan 2021 – Sep 2024

SKILLS

AI and agents Claude Code, Claude API and Platform, MCP servers, multi-agent setups, guardrails and guard tests.

Build Python and JavaScript (fundamentals); Next.js, React, Supabase. Rust and TypeScript through AI-assisted work, directed and reviewed by me.

Security Digital forensics, binary exploitation, Linux, Windows hardening, CTFs.

Spoken English, Mandarin.